

SUNSPOKE · DESIGN DECISIONS REGISTER

SunSpoke design decisions register

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DOCUMENT CONTROL

Document ID	SSP-DEC-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sunspoke		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Approved follow-ups: wet-weather brake blocks added to the BOM (line 15) and the model; wet stopping distance calculated; Value engineering section updated (bike kit USD 208); To confirm item 11 added (wet target value and block friction)
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all eleven open decisions (SSP-DDR-003 accepted); moved to decisions made; To confirm items 1 and 2 updated
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from the decision records, the review note and the design for construction

SunSpoke design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The motor's axle has 10 mm flats (or, if the donor survey shows 3/8 in slots are the norm, flats that fit them; decided 2026-10-02), an M12 thread and enough length for three threads past the nut over the dropout, torque arm and washer	The torque arms and dropouts are sized to it	SSP-DDR-003, P7
2	The donor's dropout slots take the 10 mm flats without filing (never file the fork, decided 2026-10-02), its down tube is round and 28 to 32 mm, and the step from dropout face to blade outside is about 8 mm	The saddles, torque arm slot and joggle are sized to these	SSP-DDR-003, P1 and P7
3	The donor's bottom bracket is cup-and-cone with a left lockring	The pedal-assist bracket fits under it	SSP-DDR-003, P9
4	The draw latch: adjustable hook, safety catch, 500 N working load or more, over centre with 50 N or less at its lever	It sets the V1 preload of 330 N	SSP-DDR-003, P4; SSP-CAL-001 section 8
5	The band clamp torque that gives 1.5 kN of band tension	The cradle retention margins assume it	SSP-CAL-001 section 8
6	The SwapCell receptacle's flange and hole pattern	The end stop's four holes are drawn for a 37 x 24 mm pattern	SSP-DDR-003, P3
7	The host adapter's base inserts (two, 40 mm apart) and the controller case size	The tray holes and the band clamps are sized to them	SSP-DDR-003, P6 and P8
8	The panel's mounting hole positions on its frame	The stand's rail holes are drilled to match	SSP-DDR-003, P12
9	The pack moves from legacy discharge (state 5) to heartbeat discharge (mode 2) without opening its output	The bike relies on it if the adapter is slow to start; raised with SwapCell	SSP-DDR-002, item 18
10	A low-cost boost charger with input-voltage tracking gives near-MPPT yield from one 18 V panel	R8 assumes 92 %	SSP-PRC-001, open questions
11	The value of R11's wet stopping target (proposed 14 m from 20 km/h, 1.5 times the dry target) and the wet friction of the wet-weather blocks (assumed 0.25; maker's data needed)	The wet stop of 11.9 m and the decision to keep the 20 km/h cut-off rest on both; the blocks' USD 6 a pair price is also indicative	SSP-CAL-001 section 10; bom line 15

■ Value engineering

Value-engineering target: USD 250 for the bike kit (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 208 for the bike kit (USD 42 under the target). The solar set is costed separately at USD 90, and the SwapCell pack (USD 414) is priced once in the SwapCell project. Main cost drivers and savings worth trying:

- The largest lines are the motor wheel (USD 72), the receiver cradle (USD 33), the controller (USD 25), the harness (USD 15), the host adapter (USD 14) and the handlebar display and brake sensors (USD 14).

- The wet-weather brake blocks (two pairs, USD 12, decided 2026-10-02) are the newest line.
- Making the design constructable added USD 8 to the bike kit (cradle USD 28 to 33, hardware USD 8 to 11) and USD 4 to the solar set (stand USD 10 to 14).
- Savings worth trying: buy the receptacle, draw latch and hinge in batches with other SwapCell vehicle receivers; fold the tray from 2 mm aluminium if the retention and end stop margins still hold; a shared panel and stand at a village hub removes the USD 90 solar set for riders who swap packs there.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	48 V on the SwapCell pack; budget kept at USD 250 with R12 covering the bike kit and the solar set costed separately; geared front hub; pack on the down tube; local lacing; 20 km/h default cut-off; pedal assist and walk assist, no throttle; charge on the bike; SwapCell interface v0.3 items W, C and V; pack priced once in SwapCell	Amish: go with recommendation	SSP-DDR-001, items 1 to 13
2026-09-25	Handlebar power switch in the INTERLOCK loop; motor thermistor with a controller derate instead of a cutout; raise the legacy-to-heartbeat transition with SwapCell	Amish: "i accept all your recommendations, go with them across all repos."	SSP-DDR-002, items 16 to 18
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SSP-DDR-003 (Draft, open for review: open decision 1)
2026-10-01	Treat budget_usd as a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	STANDARDS section 18; this register's Value engineering section
2026-10-02	Design for construction accepted: the changes P1 to P13 as made (open item 1)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-DDR-003, Table 1
2026-10-02	Pack swap: gate and draw latch as modelled (option a) for the prototype, reviewed after the first swap trials (open item 2)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-DDR-003, A1
2026-10-02	Panel stand in timber (option a) (open item 3)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-DDR-003, A2; review note 2026-09-26, proposal 4
2026-10-02	Donor fork slots are never filed for now: donors are chosen whose slots take 10 mm flats, and if the donor survey shows 3/8 in slots are the norm, a motor whose axle flats fit them is looked for first (open item 4)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-DDR-001, item 15
2026-10-02	Default region western Kenya and eastern Uganda, where 28 in roadsters and bicycle taxis are common, with a bicycle mechanics' group or rural transport organization there as the partner type; the partner is named when the portfolio picks partners for this area (open item 5)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-DDR-001, item 14
2026-10-02	Vertical cranks in the renders accepted (open item 6)		

Date	Decision	Decided by	Record
		Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, proposal 1
2026-10-02	Saddle raised about 12 mm in the renders accepted (open item 7)	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, proposal 2
2026-10-02	Brake sensors under the bar beside the levers accepted, as <code>model.py</code> already has them (SSP-DDR-003, P10) (open item 8)	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, proposal 3
2026-10-02	Seated rider kept in the hero render, revisited if it reads oddly (open item 9)	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, proposal 5
2026-10-02	Wet braking treated as a safety item: a wet stopping target is added to R11, wet-weather brake blocks suited to steel rims go in the kit, and trials keep a no-wet-riding rule until a wet braking test meets the target (open item 10)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-PRC-001, open questions; SSP-CAL-001 section 10; SSP-REQ-001 R11
2026-10-02	Reference motor: a widely sold 250 W front geared hub, such as one of Bafang's; its maker is asked for winding resistance, thermal capacity and gear temperature limit and R3 is rerun; the winding resistance is measured at TRL 4 if the data are not published (open item 11)	Amish: "i approve your recommendations for all 555 open decisions."	SSP-PRC-001, open questions